

2.2 Return Status

Purpose of return status is to control the printer status from the host, and the printer returns the status after receiving the status request command.

There are three types of return status; STATUS 3, STATUS 4, and STATUS 5. The return status is as follows.
The printer returns the status after receiving the status request command.

Communication Protocol	Command	Command name	Note
READY/BUSY	—	—	
XON/XOFF	—	—	
Status 3	ENQ	Status request command	
	CAN	Cancel request command	
Status 4	ENQ	Status request command	
	CAN	Cancel request command	
	DLE	Print pause request command	
	DC1	Print resume request command	
Status 5	SOH + ENQ	Status request command	
	SOH + CAN	Cancel request command	
	SOH + DLE	Print pause request command	
	SOH + DC1	Print resume request command	

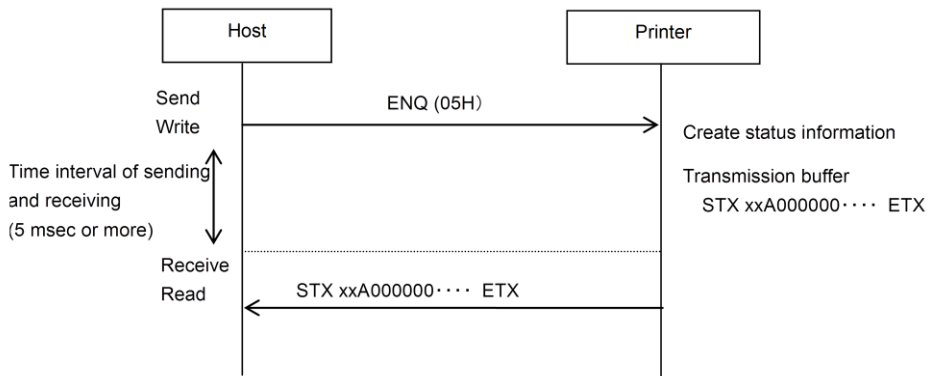
* In status 3 and 4, the printer processes the command for ESC+ENQ and ESC+CAN and then returns the status, however, for ESC+DLE and ESC+DC1, the printer neither processes the command, nor returns the status.

[Note]

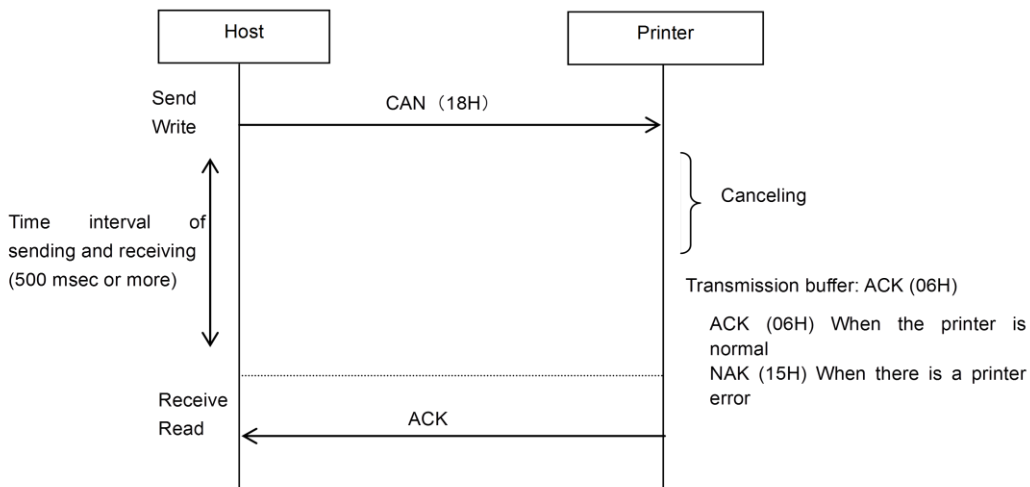
In order to obtain return status from the printer, insert the time interval of sending and receiving after sending each command to the printer.

Command	Command name	Time interval of sending and receiving (performance)
ENQ (05H)	Status request	5 msec or more * In case of wireless LAN, 300 msec or more
CAN (18H)	Cancel request	500 msec or more
DLE (10H)	Print pause request	
DC1 (11H)	Print resume request	

(1) ENQ (Status request)



(2) CAN (Cancel request)



The values stated above are reference of waiting time until the printer confirms the ACK return. Actual value varies depending on the type of interface, settings and the conditions.

Guideline of send and receive time interval of cancel request at buffer near full is more than 900 msec.

2.2.1 Return status of status 3

The printer returns the printer's status and reply as the status by receiving 3 types of request commands and print command with this communication protocol.

The details of each request command and return status are as follows.

1) Status request command

The printer returns the job ID number of receive data that is currently printing, the printer status and the number of remaining labels to print to the host by receiving this command. When the print is completed or when there is no receive data, the printer returns print quantity of all "0" (HEX 30H). If the command Job ID Number <ID> is not specified, the ID number returns space (HEX 20H).

Avoid sending ENQ while sending the print data (STX <A> - <Z> ETX). This may cause return status error or print error.

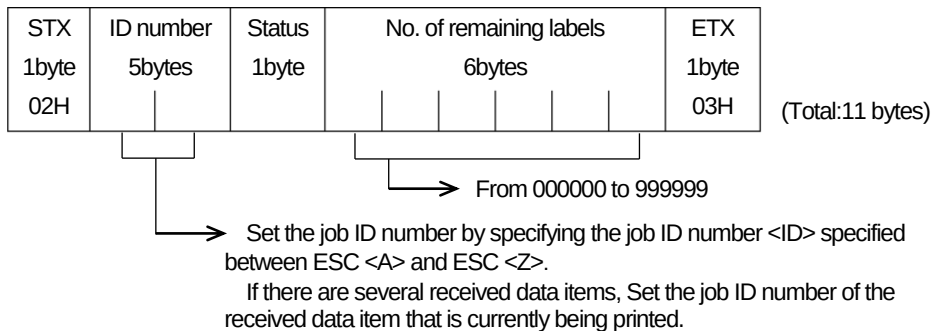
(1) Command ENQ (HEX 05H)

STX (HEX 02H)

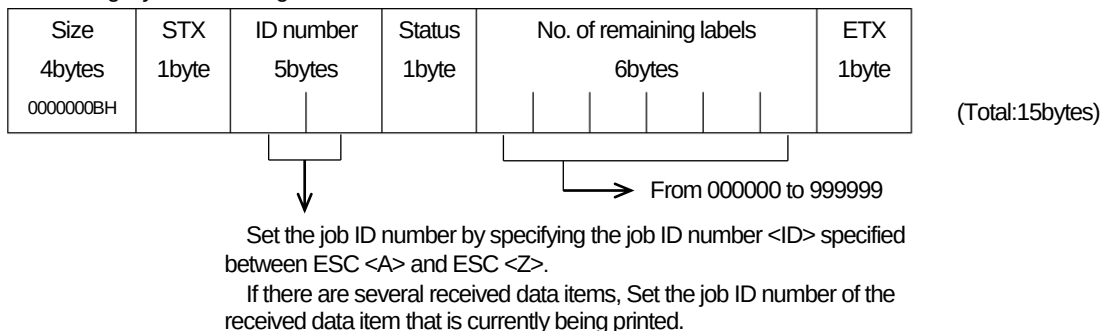
(2) Return status format

ETX (HEX 03H)

When Legacy Status setting for Port 3 is invalid 0



When Legacy Status setting for Port 3 is invalid



(3) Status list

DESCRIPTION			ASCII	HEX
OFFLINE STATE	NO ERROR		0	30
	RIBBON NEAR END		1	31
	BUFFER NEAR FULL		2	32
	RIBBON NEAR END & BUFFER NEAR FULL		3	33
	(NOT USED) BATTERY NEAR END		4	34
	(NOT USED) BATTERY NEAR END & RIBBON NEAR END		5	35
	(NOT USED) BATTERY NEAR END & BUFFER NEAR FULL		6	36
	((NOT USED) BATTERY NEAR END & RIBBON NEAR END &BUFFER NEAR FULL		7	37
ONLINE STATE	WAIT TO RECEIVE	NO ERROR	A	41
		RIBBON NEAR END	B	42
		BUFFER NEAR FULL	C	43
		RIBBON NEAR END&BUFFER NEAR FULL	D	44
		(NOT USED) BATTERY NEAR END	!	21
		(NOT USED) BATTERY NEAR END & RIBBON NEAR END	"	22
		(NOT USED)BATTERY NEAR END &BUFFER NEAR FULL	#	23
		(NOT USED) BATTERY NEAR END & RIBBON NEAR END & BUFFER NEAR FULL	\$	24
	PRINTING	NO ERROR	G	47
		RIBBON NEAR END	H	48
		BUFFER NEAR FULL	I	49
		RIBBON NEAR END & BUFFER NEAR FULL	J	4A
		(NOT USED) BATTERY NEAR END	%	25
		(NOT USED) BATTERY NEAR END & RIBBON NEAR END	&	26
		(NOT USED) BATTERY NEAR END & BUFFER NEAR FULL	'	27
		(NOT USED) BATTERY NEAR END & RIBBON NEAR END &BUFFER NEAR FULL	(	28
	STANDBY (Waiting for dispenser/cut)	NO ERROR	M	4D
		RIBBON NEAR END	N	4E
		BUFFER NEAR FULL	O	4F
		RIBBON NEAR END & BUFFER NEAR FULL	P	50
		(NOT USED) BATTERY NEAR END	)	29
		(NOT USED) BATTERY NEAR END & RIBBON NEAR END	*	2A
		(NOT USED) BATTERY NEAR END & BUFFER NEAR FULL	+	2B
		(NOT USED) BATTERY NEAR END & RIBBON NEAR END & BUFFER NEAR FULL	,	2C
	ANALYZING/ EDITING	NO ERROR	S	53
		RIBBON NEAR END	T	54
		BUFFER NEAR FULL	U	55
		RIBBON NEAR END & BUFFER NEAR FULL	V	56
		(NOT USED) BATTERY NEAR END	-	2D
		(NOT USED) BATTERY NEAR END & RIBBON NEAR END	.	2E
		(NOT USED) BATTERY NEAR END & BUFFER NEAR FULL	/	2F
		(NOT USED) BATTERY NEAR END & RIBBON NEAR END & BUFFER NEAR FULL	@	40
ERROR DETECTION	BUFFER OVER*1		a	61
	HEAD OPEN		b	62
	PAPER END		c	63
	RIBBON END		d	64
	MEDIA ERROR		e	65
	SENSOR ERROR		f	66
	BARCODE READ/MATCHING ERROR		f	66
	BARCODE READER CONNECTION CHECK ERROR		f	66
	HEAD ERROR		g	67
	(NOT USED) COVER OPEN		h	68
	CUTTER OPEN ERROR		h	68
	(NOT USED) RIBBON CORE NON-LOCK ERROR		h	68
	CARD ERROR		i	69
	CUTTER ERROR		j	6A
	OTHER ERRORS		k	6B
	(NOT USED) CUTTER SENSOR ERROR		l	6C
	(NOT USED) STACKER OR REWINDER FULL WINDER FULL		m	6D
	RFID TAG ERROR		o	6F
	RFID PROTECT ERROR		p	70
	(NOT USED) BATTERY ERROR		q	71

Depending on the timing of
analysis/editing, print quantity
may not be set up properly

Depending on the timing of analysis/editing, print quantity will not be set up properly

*1 Occurring of buffer over depends on the interfaces

RS-232C	USB	LAN	Wireless LAN	Bluetooth	IEEE1284
○	×	×	×	×	×

[○: Interface that buffer-over occurs, ×: Interface that buffer-over doesn't occur]

2) Cancel request command

This command enables to cancel print jobs and to clear the entire contents of receive buffer.

The printer returns the printer status of after completion of current process.

*When the cancel request command has sent, wait more than 500 msec before sending the next data.

*This command shall not be used during sending printer data and other data.

(1) Command CAN (HEX 18H)

(2) Return status list

Return status	Description
ACK (HEX 06H)	No error in the printer
NAK (HEX 15H)	Error in the printer

3) Print Command

The printer starts printing by receiving the print command (STX <A> ~<Z> ETX).

The printer returns the printer status after receiving process is completed.

(1)Return status list (Printer status)

RS-232C or Bluetooth

Return status	Description
ACK (HEX 06H)	No error in the printer
NAK (HEX 15H)	Error in the printer

*In version after 1.8.0, Printer returns ACK not NAK when it is in Sleep mode and Wake Up mode.

Except RS-232C and Bluetooth

Return status	Description
None	Not return status

Example of the job ID number command <ID>

```
<A>
<ID>01
<V>100<H>100<P>2<L>0202<X20>,ABC
<Q>1
<Z>
```

Refer to the "Command Specification" for the details of job ID number command <ID>.

2.2.2 Return status of status 4

The printer returns its status and reply as the status upon receiving four types of request commands and print command with this communication protocol.
Details of request commands and return status are described below.

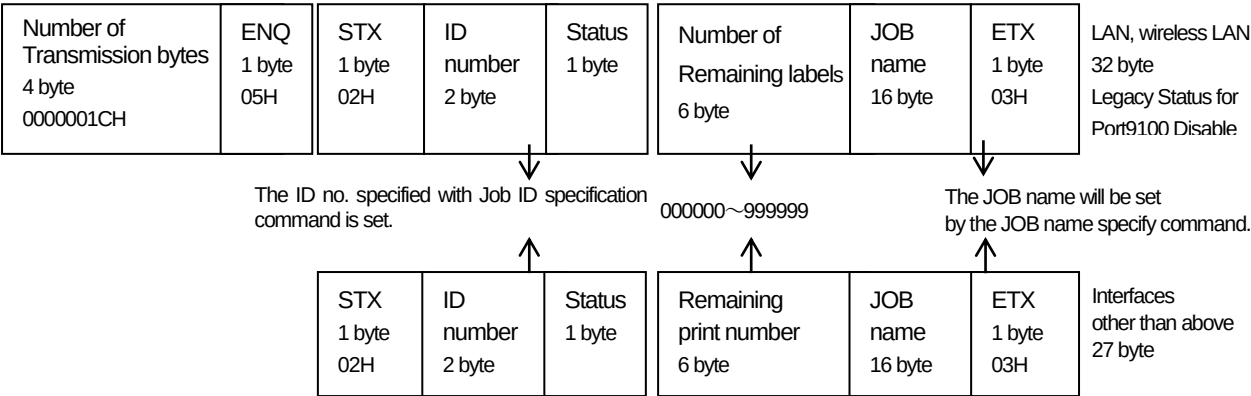
1) Status request command

The printer returns the job ID number of receive data that is currently printing, the printer status and the number of remaining labels to print to the host by receiving this command. When the print is completed or when there is no receive data, the printer returns print quantity of all "0" (HEX 30H).If the command for the designation of job ID number <ID> is not specified, the printer returns the space (HEX 20H) as the ID number.
Avoid sending ENQ while sending the print data (STX <A> - <Z> ETX). This may cause return status error or print error.

- (1) Command ENQ (HEX 05H)
- (2) Return status format

LAN, Wireless LAN 36 bytes LAN
Legacy Status setting is valid.

Number of transmission bytes 4 bytes 00000020H	Number of transmission bytes 4 bytes 0000001CH	ENQ 1byte 05H	STX 1byte 02H	ID No. 2bytes	Status 1byte	No. of Remaining labels 6bytes	JOB name 16bytes	ETX 1byte 03H
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(3) Return status list

DESCRIPTION			ASCII	HEX
OFFLINE STATE	NO ERROR		0	30
	RIBBON NEAR END		1	31
	BUFFER NEAR FULL		2	32
	RIBBON NEAR END & BUFFER NEAR FULL		3	33
	PRINT HALT (NO ERROR)		4	34
	(NOT USED) BATTERY NEAR END		5	35
	(NOT USED) BATTERY NEAR END & RIBBON NEAR END		6	36
	(NOT USED) BATTERY NEAR END & BUFFER NEAR FULL		7	37
(NOT USED) BATTERY NEAR END & RIBBON NEAR END&BUFFER NEAR FULL		8	38	
ONLINE STATE	WAIT TO RECEIVE	NO ERROR	A	41
		RIBBON NEAR END	B	42
		BUFFER NEAR FULL	C	43
		RIBBON NEAR END & BUFFER NEAR FULL	D	44
		PRINT HALT (NO ERROR)	E	45
		(NOT USED) BATTERY NEAR END	!	21
		(NOT USED) BATTERY NEAR END & RIBBON NEAR END	"	22
		(NOT USED) BATTERY NEAR END & BUFFER NEAR FULL	#	23
	(NOT USED) BATTERY NEAR END & RIBBON NEAR END &BUFFER NEAR FULL	\$	24	
	PRINTING	NO ERROR	G	47
		RIBBON NEAR END	H	48
		BUFFER NEAR FULL	I	49
		RIBBON NEAR END & BUFFER NEAR FULL	J	4A
		PRINT HALT (NO ERROR)	K	4B
		(NOT USED) BATTERY NEAR END	%	25
		(NOT USED) BATTERY NEAR END & RIBBON NEAR END	&	26
		(NOT USED) BATTERY NEAR END & BUFFER NEAR FULL	'	27
	(NOT USED) BATTERY NEAR END & RIBBON NEAR END &BUFFER NEAR FULL	(	28	
	STANDBY (waiting for dispenser/cutter)	NO ERROR	M	4D
		RIBBON NEAR END	N	4E
		BUFFER NEAR FULL	O	4F
		RIBBON NEAR END & BUFFER NEAR FULL	P	50
		PRINT HALT (NO ERROR)	Q	51
		(NOT USED) BATTERY NEAR END	)	29
		(NOT USED) BATTERY NEAR END & RIBBON NEAR END	*	2A
		(NOT USED) BATTERY NEAR END & BUFFER NEAR FULL	+	2B
	(NOT USED) BATTERY NEAR END & RIBBON NEAR END &BUFFER NEAR FULL	,	2C	
	ANALYZING/ EDITING	NO ERROR	S	53
		RIBBON NEAR END	T	54
		BUFFER NEAR FULL	U	55
		RIBBON NEAR END & BUFFER NEAR FULL	V	56
		PRINT HALT (NO ERROR)	W	57
		(NOT USED) BATTERY NEAR END	-	2D
		(NOT USED) BATTERY NEAR END & RIBBON NEAR END	.	2E
		(NOT USED) BATTERY NEAR END & BUFFER NEAR FULL	/	2F
	(NOT USED) BATTERY NEAR END & RIBBON NEAR END &BUFFER NEAR FULL	@	40	
Error detection	HEAD OPEN		b	62
	PAPER END		c	63
	RIBBON END		d	64
	MEDIA ERROR		e	65
	SENSOR ERROR		f	66
	BARCODE READ/MATCHING ERROR		f	66
	BARCODE READER CONNECTION CHECK ERROR		f	66
	HEAD ERROR		g	67
	(NOT USED) COVER OPEN		h	68
	CUTTER OPEN ERROR		h	68
	(NOT USED) RIBBON CORE NON-LOCK ERROR		h	68
	CARD ERROR		i	69
	CUTTER ERROR		j	6A
	OTHER ERRORS		k	6B
	(NOT USED) CUTTER SENSOR ERROR		l	6C
	(NOT USED) STACKER OR REWINDER FULL WINDER FULL		m	6D
	RFID TAG ERROR		o	6F
	RFID PROTECT ERROR		p	70
	(NOT USED) BATTERY ERROR		q	71

Depending on the timing of analysis/editing, print quantity will not be set up properly

*1 Buffer over may occur depending on interface

RS-232C	USB	LAN	Wireless LAN	Bluetooth	IEEE1284
○	×	×	×	×	×

[○: Interface that buffer-over occurs, ×: Interface that buffer-over doesn't occur]

- **Cause of receive buffer near full**

Buffer near full occurs when remaining capacity of receive buffer (2.95MB) becomes 0.95 MB.

- **Factors to release receive buffer near full**

Receive buffer near full is released when remaining capacity of the receive buffer becomes 1.95 MB or more.

2) Cancel request command

This command enables to cancel print jobs and to clear the entire contents of receive buffer.

The printer returns the printer status of after completion of current process.

*When the cancel request command has sent, wait more than 500 msec before sending the next data.

Do not send cancel request command within STX <A>~<Z> ETX. CAN (cancel request) may not be send correctly.

(1) Command CAN (HEX 18H)

(2) Return status list

Return Status	Description
ACK (HEX 06H)	No error in the printer
NAK (HEX 15H)	Error in the printer

*In the case of LAN / wireless LAN interface, ACK hexadecimal 0000000106H, NAK will be hexadecimal 0000000115H on the condition that Legacy status is invalid.

3) Print Command

Print command (<A>~<Z>) starts the printing process.

The printer returns the printer status after receiving process is completed.

(1) Return status list

RS-232C or Bluetooth

Return Status	Description
ACK (HEX 06H)	No error in the printer
NAK (HEX 15H)	Error in the printer

*In version after 1.6.4, Printer returns ACK not NAK when it is in Sleep mode and Wake Up mode.

Except RS-232C and Bluetooth

Return Status	Description
None	Not return status

4) Print pause request command

This command stops the printing process.

The printer returns the printer status after receiving process is completed.

(1) Command DLE (HEX 10H)

(2) Return status list

Return Status	Description
ACK (HEX 06H)	No error in the printer
NAK (HEX 15H)	Error in the printer

* In the case of LAN / wireless LAN interface, ACK hexadecimal 0000000106H, NAK will be hexadecimal 0000000115H

Do not send print pause request command within STX <A>~<Z> ETX. DLE (print pause request) may not be send correctly.
DLE (print pause request) in font data, graphics data and barcode data will not be processed.

5) Print resume request command

This command enables to release the print pause and resume printing.

The printer returns the printer status after receiving process is completed.

(1) Command DC1 (HEX 11H)

(2) Return status list

Return Status	Description
ACK (HEX 06H)	No error in the printer
NAK (HEX 15H)	Error in the printer

* In the case of LAN / wireless LAN interface, ACK hexadecimal 0000000106H, NAK will be hexadecimal 0000000115H.

2.2.3 Return status of status 5

The functions in this communication protocol are as follows.

1)Data transmission format

- (1) Control commands such as print data, status request, cancel request and so on need to be within STX (HEX 02H) and ETX (HEX 03H).
* No need to add STX (HEX 02H) and ETX (HEX 03H) for setting data.
- (2) Item No. should be added to the print data.
- (3) When BCC check function is enabled, it is necessary to put BCC at the end of item (<Z>).
However, there is no need for putting BCC at the end of information acquiring command such as status request.

Examples)

Print Data

STX	<A>	Print Data	<Z>	BCC	ETX
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Status request

STX 1 byte 02H	SOH 1 byte 01H	ENQ 1 byte 05H	Item number 00000~99999 or *****	ETX 1 byte 05H
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2) Item Status Request Command

Upon receive of this command, the printer will search the specified item No. from the end of the history buffer, and return the status of appropriate item.

[Command]

SOH (HEX 01H) + ENQ (HEX 05H)

[Format]

STX + SOH + ENQ + aaaaa + ETX

Parameter

a "Item No." Valid range: 00000 to 99999

1. The parameter should be always 5 digits.
2. The parameter "*" is "HEX 2AH".
3. The item number is used for compatibility with the conventional models, and the specification of the item number is ignored.
Always the status of the last item is returned.

[Return status format] (Legacy Status setting is valid with LAN/Wireless LAN, Other I/F)

[1] STX 02H	[5] Specified item No. *1	[2] Specified item status *1	[5] Item No.in process	[2] Item status in process *2	[6] Print quantity of item in process	[1] ETX 03H
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LANW-LAN Legacy Status for Port9100 Enable

[4] Size	[1] STX 02H	[5] Specified item No. *1	[2] Specified item status *1	[5] Item No.in process	[2] Item status in process *2	[6] Print quantity of item in process	[1] ETX 03H
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Number in [] shows number of byte, and return status format consists of 22 byte (fix). Item No. is filled by "0".
Item No. in process becomes space (20H) when print completed.

*1 Specified item number is ***** (0x2A2A2A2A2A) and the last item status is fixed ** (0x2A2A).

*2 The first digit is the status and the second digit is the error in the currently processed item status.

The below is the currently processed item status list.

Digit position	Description	ASCII	HEX
1	Offline	0	30
	Online, waiting for receive	1	31
	Online, printing	2	32
	Online, standby (waiting for dispensing)	3	33
	Online, analyzing/editing	4	34
	* The "number of currently processed item" may not be set due to the timing.		
	Error	5	35
	Print pause (when received DLE command)	6	36

Off line	Off line	ASCII	HEX
2	No error	0	30
	Ribbon near end	1	31
	Buffer near full	2	32
	Ribbon near end & Buffer near full	3	33
	Label near end	4	34
	Label near end & Ribbon near end	5	35
	Label near end & Buffer near full	6	36
	Label near end & Ribbon near end & Buffer near full	7	37
	Machine error	A	41
	Flash ROM error	B	42
	Head open	C	43
	Paper end	D	44
	Ribbon end	E	45
	Sensor error	F	46
	Head error	G	47
	Memory read/write error (Cartridge read/write error)	H	48
	Memory full(Cartridge full)	I	49
	Cutter error	J	4A
	Calendar error	K	4B
	Kanji data error (Kanji ROM error)	L	4C
	(Not used) Internal rewinder full	M	4D
	(Not used) Cutter bracket open error	N	4E
	RFID tag error	O	4F
	RFID protect error	P	50
	BCC error	Q	51
	(Not used) Item No. error	R	52
	Media error	S	53
	(Not used) Ribbon winding non lock error	T	54
	Other error	U	55
	(Not used) Cover open	V	56
	Barcode read/matching error	V	56
	Barcode reader connection check error	W	57

[Examples]

- (1) STX SOH ENQ 00001 ETX
- (2) STX SOH ENQ ***** ETX

[Supplemental explanation]

1. This command shall not be used during sending other data such as print data. The status may not be returned or print may not be finished correctly when sending this command during print data sending.
2. The printer returns the current status after receiving this command.